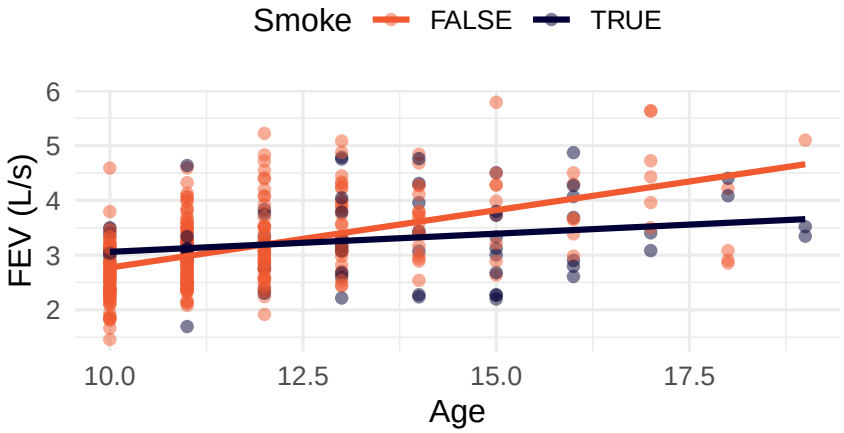
# Day 12: Inference for linear combinations

We're interested in modeling **lung capacity** (FEV) on **age** and **smoking status** for a sample of adolescents. As kids get older, FEV generally increases, but we also think FEV decreases for smokers. Since smokers are likely to be older, we need a model that accounts for both.




Theoretical mean model:

$$\mu\{FEV \mid Age, Smoke\} = \beta_0 + \beta_1\, Age + \beta_2\, Smoke + \beta_3\, Age \times Smoke$$

| term | estimate | std. error | t | p-value |
|---|---|---|---|---|
| Intercept | 0.675 | 0.251 | 2.68 | 0.008 |
| Age | 0.210 | 0.021 | 9.97 | <0.001 |
| SmokeTRUE | 1.720 | 0.563 | 3.05 | 0.002 |
| Age:SmokeTRUE | -0.143 | 0.042 | -3.40 | <0.001 |

**Example:** Models for each group

Nonsmokers ($Smoke = 0$): $\mu\{FEV \mid Age\} =$

$\hat{y} =$

Smokers ($Smoke = 1$): $\mu\{FEV \mid Age\} =$

$\hat{y} =$

## Reading the table: what is significantly different than 0?

- The intercept for non-smokers is _____
- The intercept offset for smokers is _____
- The slope for non-smokers is _____
- The *difference* in slopes between smokers and non-smokers is _____
- The slope for smokers is _____

The slope for smokers is $\beta_1 + \beta_3$. The table gives us $SE(\hat{\beta}_1)$ and $SE(\hat{\beta}_3)$, but **not** the standard error of their sum.

**Why can't we just add $SE(\hat{\beta}_1) + SE(\hat{\beta}_3)$?**

**Why can't we use the p-value for** `Age:SmokeTRUE` **to test whether the smoker slope is zero?**

> **Definition**
>
> A **linear combination** of coefficients has the form
>
> $$\gamma = c_1\beta_i + c_2\beta_j$$
>
> where $c_1$ and $c_2$ are constants
> **Estimate:** $\hat{\gamma} =$ _____
> **Standard error:**
>
> $$SE(\hat{\gamma}) = \sqrt{c_1^2\{SE(\hat{\beta}_i)\}^2 + c_2^2\{SE(\hat{\beta}_j)\}^2 + 2c_1c_2\,Cov(\hat{\beta}_i, \hat{\beta}_j)}$$

The extra term is there because $\hat{\beta}_i$ and $\hat{\beta}_j$ come from the *same* fitted model, so their estimation errors are not independent. $Cov(\hat{\beta}_i, \hat{\beta}_j)$ measures how they move together.

## The covariance matrix of the coefficients

```
vcov(fev_lm) |> round(4)
```

```
                (Intercept)     Age SmokeTRUE Age:SmokeTRUE
(Intercept)          0.0632 -0.0052   -0.0632         0.0052
Age                 -0.0052  0.0004    0.0052        -0.0004
SmokeTRUE           -0.0632  0.0052    0.3172        -0.0234
Age:SmokeTRUE        0.0052 -0.0004   -0.0234         0.0018
```

The **diagonal** entries are the variances, $V(\hat{\beta}_j) = \{SE(\hat{\beta}_j)\}^2$.

- Check: $SE(\hat{\beta}_1) =$ _____ , so $\{SE(\hat{\beta}_1)\}^2 =$ _____

The **off-diagonal** entries are the covariances, $Cov(\hat{\beta}_i, \hat{\beta}_j)$. - The matrix is symmetric: $Cov(\hat{\beta}_i, \hat{\beta}_j) = Cov(\hat{\beta}_j, \hat{\beta}_i)$.

## Example 1: CI for the smoker slope

**Target:** $\gamma = \beta_1 + \beta_3$. So $c_1 =$_____ and $c_2 =$_____

**Estimate:**

**Standard error:** pull the pieces out of the covariance matrix.

| piece | value |
|---|---|
| $\{SE(\hat{\beta}_1)\}^2$ | |
| $\{SE(\hat{\beta}_3)\}^2$ | |
| $Cov(\hat{\beta}_1, \hat{\beta}_3)$ | |

**90% CI:** $\hat{\gamma} \pm t^*_{n-(p+1)} \cdot SE(\hat{\gamma})$, where $n = 345, p = 3$, so $df = 341$ and $t^* \approx 1.649$.

**Interpretation:**

> **What if we ignored the covariance?**
>
> Using only $\sqrt{\{SE(\hat{\beta}_1)\}^2 + \{SE(\hat{\beta}_3)\}^2}$ would give 0.047. The correct SE is 0.037. Here the covariance is **negative**, so ignoring it makes the interval too wide.

## Testing a linear combination

**Hypotheses:** $H_0 : \gamma = \gamma_0$ vs. $H_a : \gamma \neq, <, \text{or} > \gamma_0$

**Test statistic:** $t = \dfrac{\hat{\gamma} - \gamma_0}{SE(\hat{\gamma})}$

**Reference distribution:** $t$ distribution with $df = n - (p + 1)$

**p-value:** area in the tail(s) specified by $H_a$

Same four steps as a test for a single coefficient. Only the estimate and the SE change.

**Example:** Is the smoker slope different from zero? Using $\hat{\gamma} = 0.0664$ and $SE(\hat{\gamma}) = 0.0366$:

## Your turn: smokers vs. nonsmokers at age 15

At age 15, how different is the mean FEV of smokers and nonsmokers?

**0.** Look back and the graph and identify what this question is asking about. What do you think you will find?

**1.** Write the theoretical mean function for each group evaluated at $Age = 15$, and subtract (smoker $-$ nonsmoker) to find $\gamma$.

**2.** The difference is $\gamma =$_____, so $c_2 =$ _____ $c_3 =$ _____

**3.** Find $\hat{\gamma}$

**3.** Write out $SE(\hat{\gamma})$ using the values from the covariance matrix.

**4.** Construct a 95% CI for $\gamma$ ($df = 341, t^* \approx 1.967$).

**5.** Test $H_0 : \gamma = 0$ vs. $H_a : \gamma \neq 0$.

**6.** Interpret your results in context.

## A shortcut: re-center the model

If we measure age as "years since age 15", the coefficient on Smoke *is* the smoker-minus-nonsmoker difference at age 15, and the standard table gives us its SE directly:

```
fev <- mutate(fev, Age15 = Age - 15)
lm(FEV ~ Age15 * Smoke, data = fev) |> tidy()
```

```
# A tibble: 4 x 5
  term            estimate std.error statistic   p.value
  <chr>              <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)        3.82     0.0780     49.0  2.33e-156
2 Age15              0.210    0.0210      9.97 1.01e- 20
3 SmokeTRUE         -0.430    0.125      -3.44 6.57e-  4
4 Age15:SmokeTRUE   -0.143    0.0422     -3.40 7.63e-  4
```

Compare the SmokeTRUE row to the output from the previous section.

## Summary

| To do inference for… | Use |
|---|---|
| a single coefficient $\beta_j$ | $t$-test / CI from the coefficient table |
| a linear combination $c_1\beta_i + c_2\beta_j$ | $t$-test / CI with $SE = \sqrt{c_1^2\{SE(\hat{\beta}_i)\}^2 + c_2^2\{SE(\hat{\beta}_j)\}^2 + 2c_1c_2\,Cov(\hat{\beta}_i, \hat{\beta}_j)}$ |
| several coefficients at once ($\beta_2 = \beta_3 = 0$) | extra sums of squares $F$-test |

- Never add standard errors. Use the covariance matrix.
- The reference distribution is always $t_{n-(p+1)}$ for a single linear combination.
- A linear combination is *one* number, so we use a $t$ procedure. When $d > 1$ coefficients are tested at the same time, that's an $F$-test.

## Your turn: in R

The R activity is posted on moodle. Here are some functions that might be helpful:

| Function | Purpose |
|---|---|
| `coef()` | Extract the estimated coefficients from a fitted model (as a named vector) |
| `vcov()` | Extract the covariance matrix of the estimated coefficients from a fitted model |
| `x[i]`, `m[i, j]` | Pull off element `i` of a vector, or the entry in row `i` and column `j` of a matrix |
| `sqrt()` | Take the square root (e.g., to turn a variance into a standard error) |